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Lecture Notes on Differential Equations and Transform Techniques

Spring 2025

The purpose of these lecture notes is to provide a concise summary of the key concepts covered in the course. They are designed to serve as a **quick reference** and should be used in conjunction with the core textbooks listed below. While some sections are adapted from these referenced texts, the notes are not intended to replace the original works, which offer more comprehensive explanations. To gain a deeper understanding, students are encouraged to refer to the textbooks listed.

References:

1. W. E. Boyce, R. C. DiPrima, D. B. Meade, *Elementary Differential Equations and Boundary Value Problems*, 9th edition, Wiley, 1997.

2. Brad G. Osgood, *Lectures on Fourier Transform and Its Applications*, IAS/Park City Mathematics Series / American Mathematical Society (AMS), 2019.

3. Charles Henry Edwards, David E. Penney, *Differential Equations and Boundary Value Problems: Computing and Modeling*, 4th edition, Pearson, 2000.

4. George F. Simmons, *Differential Equations with Applications and Historical Notes*, 2nd edition, CRC Press, 2016.

5. Ruel V. Churchill, *Fourier Series and Boundary Value Problems*, McGraw-Hill, 1941.

6. Gustav Doetsch, *Introduction to the Theory and Application of the Laplace Transformation*, Springer, 2012.

7. Morris Tenenbaum, Harry Pollard, *Ordinary Differential Equations*, Dover Books on Mathematics, Dover Publications, 1985.

EE1060: Differential Equations and Transform Techniques

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Topics covered :

1. General solution of Second-order nonhomogeneous ODE
2. Idea of Annihilation

- General Solution of nonhomogeneous ODE: Let $y_p(x)$ denote any particular solution to the ODE $L[y] = g(x)$ and $\{y_1(x), \cdots, y_n(x)\}$ denote the fundamental solution for the homogeneous part i.e., $L[y] = 0$. Then, the general solution for the nonhomogeneous DE is given by

$$y(x) = \underbrace{\sum_{i=1}^{n} c_i y_i(x)}_{y_c(x)} + y_p(x) \tag{1}$$

where $y_c(x)$ is the complementary solution.

Proof: Let $y(x)$ denote any solution to the ODE $L[y] = g(x)$. Then,

$$L[y] - L[y_p] = g(x) - g(x) = 0 \implies L[y - y_p] = 0 \text{ by linearity of } L \tag{2}$$

Since, $L[y - y_p] = 0$ is a homogeneous ODE, its general solution can be expressed as a linear combination of the the fundamental solutions i.e.,

$$y(x) - y_p(x) = \sum_{i=1}^{n} c_i y_i(x) = y_c(x) \implies y(x) = \sum_{i=1}^{n} c_i y_i(x) + y_p(x) \tag{3}$$

- **Note on linearity of** $L[y] = g(x)$: If $y_1(x)$ is a solution to $L[y] = g_1(x)$ and $y_2(x)$ is a solution to $L[y] = g_2(x)$, then $\alpha y_1(x) + \beta y_2(x)$ is a solution to $L[y] = \alpha g_1(x) + \beta g_2(x)$.

- An operator L is an Annihilator of $g(x)$ if $L[g(x)] = 0$. In the context of differential equations, Annihilators are built using derivatives or the differential operator D. For example, consider a second-order homogeneous ODE with constant coefficients

$$\frac{d^2y}{dx^2} + p\frac{dy}{dx} + qy = 0 \tag{4}$$

In terms of differential operator, (4) can be written as

$$\left(D^2 + pD + q\right) y = 0 \tag{5}$$

So, for any solution $y(x)$ of (4), the operator $L(D) = \left(D^2 + pD + q\right)$ is an Annihilator.

- The Annihilators for some common functions[1] are

[1] Recall teh arguments covered in the class

1. For $g(x) = a_n x^n + \cdots + a_1 x + a_0$, the annihilator is $L(D) = D^{n+1}$.
2. For $g(x) = e^{ax}$, the annihilator is $L(D) = (D - a)$.
3. For $g(x) = x^n e^{ax}$, the annihilator is $L(D) = (D - a)^{n+1}$.
4. For $g(x) = a_n x^n e^{ax} + \cdots + a_1 x e^{ax} + a_0 e^{ax}$, the annihilator is $L(D) = (D - a)^{n+1}$.
5. For $g(x) = \cos mx$ and $g(x) = \sin mx$, the annihilator is $L(D) = \left(D^2 + m^2\right)$.
6. For $g(x) = a_n x^n e^{ax} \cos mx + \cdots + a_1 x e^{ax} \cos mx + a_0 e^{ax} \cos mx$, the annihilator is $L(D) = \left((D - a)^2 + m^2\right)^{n+1}$.